AURORA STM32 Pinout and Communication Reference

Final Proteus embedded wiring reference

BMS STM32F401RE

Pin	Peripheral / Signal	Function
PA0	ADC1_IN0	Battery-temperature potentiometer input.
PA2	USART2_TX @ 9600	BMS state to VCU.
PA9	USART1_TX @ 115200	BMS Virtual Terminal output.
PB0	GPIO output	Green status LED.
PB1	GPIO output	Red status LED.

VCU STM32F401RE

Pin	Peripheral / Signal	Function
PA0	ADC1	Accelerator input (ACC).
PA1	ADC1	Brake input (BRK).
PA7	ADC1	SOC input.
PA4	GPIO input	Ignition / start switch.
PA5	GPIO input	Charge switch.
PA3	USART2_RX @ 9600	Receives BMS state.
PA9	USART1_TX @ 9600	Sends command to Motor Controller.
PA10	USART1_RX @ 9600	Receives Motor Controller health feedback.
PC6	USART6_TX @ 115200	VCU Virtual Terminal output.

Motor Controller STM32F401RE

Pin	Peripheral / Signal	Function
PA2	USART2_TX @ 9600	Motor health feedback to VCU.
PA3	USART2_RX @ 9600	Receives VCU motor command.
PA4	GPIO input	Motor-fault switch/input.
PA6	TIM3_CH1	Hardware PWM to MOSFET gate.
PA9	USART1_TX @ 115200	Motor Controller Virtual Terminal output.

Communication Protocols

Link	Baud	Protocol	Meaning
BMS → VCU	9600	Single byte: N / W / D / S	Normal / Warning / Derate / Shutdown.
VCU → Motor	9600	Two bytes: D/R/S + value	Drive, Regen or Stop plus command value 0–100.
Motor → VCU	9600	Single byte: O / F	Motor OK / Motor hardware fault.
Debug screens	115200	Human-readable text	Status only; separate from machine links.

Grounding

All STM32 logic sections and the motor-driver stage use a common ground reference in the current Proteus prototype. The ground symbols represent the same 0 V node; a long physical ground wire is not required when the same global GND net is used.